

Biometric Identification Through Auditory EEG Signatures

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Abstract—Intelligent authentication systems for real-world deployment must be robust to noise, scalable to new users, and capable of operating under open-set conditions. However, many existing approaches remain constrained by closed-set assumptions and limited resilience to signal degradation, reducing their practical applicability. This research presents a noise-resilient open-set authentication framework that addresses both adaptive signal enhancement and discriminative representation learning. A WaveNet-based denoising module, optimized using a scale-invariant signal-to-noise ratio objective, mitigates environmental and physiological noise. Subsequently, a deep metric learning module built on residual network backbones and angular margin-based losses learns a compact embedding space for reliable identity discrimination and unseen-user detection. The proposed framework is evaluated on the Auditory Evoked Potential Electroencephalography–Biometric Dataset, which comprises 20 participants across 12 recording sessions and auditory stimuli including native-language speech, foreign-language speech, and neutral sounds. Compared with LSTM, BiLSTM, and single-stage ResNet baselines under Gaussian, power-line, and electromyographic noise, the two-stage approach achieves up to 98.15% Precision@1 and 95.64% true acceptance rate at 1% false acceptance rate, with the largest retrieval gain of 10.52 percentage points under electromyographic noise. Further analysis highlights the benefit of explicitly decoupling noise mitigation and embedding learning objectives in intelligent authentication systems.

Index Terms—Biometric identification, EEG authentication, Open-set recognition, Deep metric learning, Signal enhancement.

I. INTRODUCTION

The rapid proliferation of intelligent systems in security-sensitive applications has intensified the demand for authentication mechanisms that are not only accurate but also robust, scalable, and adaptable to real-world conditions. Modern authentication systems are increasingly deployed in dynamic environments such as smart devices, wearable technologies, and cyber-physical systems, where signal quality, user populations, and operating conditions cannot be strictly controlled. As a

result, intelligent authentication solutions must handle noisy inputs, accommodate newly encountered users, and operate reliably in open-set scenarios [1]–[3].

Traditional authentication approaches, including password-based methods and conventional biometric systems, face well-known limitations, such as security vulnerabilities, spoofing, and limited adaptability. While recent advances in machine learning and deep learning have significantly improved biometric authentication performance, many existing methods remain constrained by closed-set assumptions, under which all user identities are assumed to be known during training [1]–[3]. Such assumptions are unrealistic in practical deployments, where systems must continuously handle previously unseen users. Moreover, real-world sensory data are often corrupted by environmental and physiological noise, resulting in degraded performance and reduced reliability [1], [4].

To address these challenges, representation learning and deep metric learning have emerged as powerful paradigms for intelligent authentication. By learning discriminative embedding spaces that preserve semantic similarity, metric learning enables identity verification, retrieval, and open-set recognition within a unified framework [5]–[8]. However, the effectiveness of embedding-based authentication systems is highly sensitive to the quality of the input signal. In noisy conditions, poorly structured embeddings can significantly impair both discrimination accuracy and unseen-user detection, underscoring the need for robust signal enhancement mechanisms [1], [4].

In parallel, recent advances in deep neural architectures for signal enhancement have demonstrated strong potential to mitigate noise in complex, non-stationary signals. Models such as WaveNet have shown remarkable ability to learn signal structures directly from raw data, enabling adaptive denoising across diverse noise conditions [13]. Despite these advances, integrating noise-adaptive signal enhancement with discriminative representation learning for open-set authentication remains underexplored, particularly from a system-level perspective [1], [2].

This research addresses these gaps by proposing an intel-

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ligent authentication framework that considers both noise resilience and open-set recognition. The framework integrates a WaveNet-based adaptive denoising module with a deep metric-learning architecture built on ResNet backbones and angular-margin-based loss functions [7], [8], [11]. By combining noise-aware signal enhancement with embedding-level discrimination, the proposed system is designed to operate reliably in realistic environments and scale to previously unseen users [1], [5], [6]. Using brain-derived signals as a representative case study, experiments on the Auditory Evoked Potential EEG–Biometric Dataset (20 subjects, 12 sessions) demonstrate that the proposed approach achieves high robustness and strong open-set recognition performance [1], [2], [4]. The results highlight the benefit of decoupling noise mitigation and representation learning objectives when designing deployable intelligent authentication systems [7], [8].

The main contributions of this work are as follows:

- A two-stage framework that explicitly decouples noise-adaptive signal enhancement from discriminative representation learning, enabling targeted optimization of each objective independently.
- Integration of a WaveNet-based denoiser with a scale-invariant SNR objective that adapts to diverse noise types (Gaussian, power-line, EMG) through learned dilated causal convolution filters.
- A comprehensive evaluation under realistic noise conditions, demonstrating higher closed-set retrieval accuracy than single-stage baselines (LSTM, BiLSTM, ResNet-34) and comparable-to-improved open-set verification across three noise types.

II. RELATED WORK

Intelligent authentication systems have attracted growing attention as security requirements increase in dynamic and resource-constrained environments such as smart devices, wearable platforms, and cyber-physical systems. Traditional authentication mechanisms, including passwords and token-based methods, are prone to well-known security vulnerabilities. At the same time, conventional biometric systems based on facial images, fingerprints, or voice signals remain susceptible to spoofing attacks [1]. As a result, learning-based authentication approaches have been widely investigated to enhance robustness and adaptability.

Early learning-based authentication studies primarily focused on closed-set classification, assuming that all user identities are known during training. Using neural or physiological signals, Zhang *et al.* employed attention-based recurrent neural networks to achieve high identification accuracy under controlled conditions [3]. Subsequent works extended this line of work by applying convolutional and Siamese network architectures to raw signal representations, demonstrating competitive performance while reducing reliance on handcrafted features [1], [2]. However, most of these approaches remain constrained by closed-set assumptions, limiting their scalability in real-world deployments.

Specifically, Fallahi *et al.* [1] proposed BrainNet, which leverages Siamese Networks for brainwave-based biometric recognition, achieving improved generalization through pairwise similarity learning. Yang *et al.* [2] investigated triplet loss-based deep learning for EEG authentication, demonstrating the benefit of metric-based training over softmax classification. Zhang *et al.* [3] introduced MindID, an attention-based recurrent neural network for person identification from brain waves. While these methods report promising results, they typically operate under closed-set assumptions and do not incorporate explicit noise mitigation stages, leaving a gap in practical robustness.

To address this limitation, open-set recognition has been increasingly explored in authentication systems. Metric learning has emerged as an effective paradigm for open-set authentication by learning embedding spaces in which identity similarity is preserved through distance relationships rather than fixed class assignments. Loss functions such as contrastive loss [5] and triplet loss [6] have been widely adopted to improve discriminative representation learning. More recently, angular margin-based losses, including ArcFace, have demonstrated superior embedding compactness and inter-class separability, leading to improved stability in identity discrimination and retrieval tasks [7]. Nevertheless, the effectiveness of metric learning-based authentication remains highly sensitive to input noise and signal variability.

Robustness to noise is a critical challenge in practical authentication systems, as real-world signals are often corrupted by environmental interference and physiological artifacts [4]. Traditional filtering and handcrafted feature extraction methods have been widely used to mitigate noise, but their performance is limited under complex, non-stationary conditions. Recent deep learning-based denoising approaches, particularly those capable of modeling temporal dependencies, have shown strong potential for adaptive noise suppression. Despite these advances, most existing studies treat signal enhancement and representation learning as independent components, without explicitly optimizing their interaction for open-set authentication.

Brain-derived signals have also been explored as an alternative authentication modality due to their inherent uniqueness and resistance to spoofing. Prior studies have reported promising identification performance using both traditional machine learning and deep learning models [1]–[3]. However, these systems remain vulnerable to noise, session variability, and deployment constraints, and open-set operation is often insufficiently addressed [4]. Additionally, ethical considerations around privacy and user consent are crucial when deploying brain-derived signal systems. Acknowledging these concerns can build trust with users by ensuring their data is handled with care, transparency, and in compliance with relevant regulations. By addressing these ethical aspects, the potential of brain-derived signals as a secure and reliable modality can be fully realized, anticipating stakeholders’ concerns while reinforcing their value in authentication.

In summary, although prior work has demonstrated the effec-

tiveness of metric learning and noise suppression for authentication, there remains a lack of integrated frameworks that jointly address noise resilience, discriminative representation learning, and open-set recognition at the system level. This gap motivates the proposed approach. Unlike prior EEG biometric studies that rely on single-stage classification models, this work formulates EEG authentication as a two-stage learning problem, decoupling deep representation learning from identity discrimination via metric learning [5]–[8]. This design enables robust open-set authentication and superior generalization to unseen subjects.

III. METHODOLOGY

This study proposes a two-stage authentication framework that enhances robustness and open-set capability by explicitly decoupling noise mitigation from representation learning. The overall architecture consists of a noise-adaptive signal enhancement stage followed by a deep metric learning–based embedding stage, as illustrated in Fig. 1.

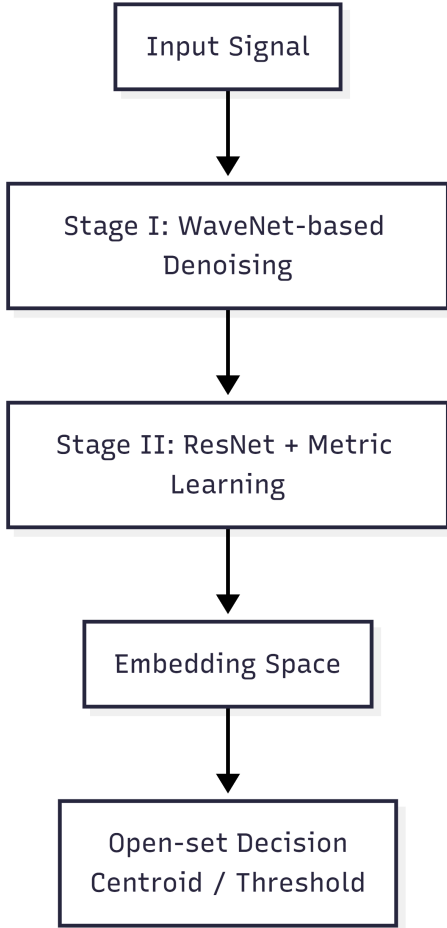


Fig. 1. Overview of the proposed two-stage authentication framework.

A. Stage I: Noise-Adaptive Signal Enhancement

The first stage aims to mitigate environmental and physiological noise that commonly degrades real-world signals [4]. A

WaveNet-based denoising model is employed to learn a direct mapping from noisy inputs to cleaner signal representations [13]. The model is trained using a scale-invariant signal-to-noise ratio (SI-SNR) objective, which encourages faithful signal reconstruction while remaining insensitive to amplitude variations. By explicitly performing denoising before representation learning, this stage reduces noise-induced distortions and improves signal consistency across sessions and operating conditions. The WaveNet denoiser architecture is shown in Fig. 2.

Adaptive mechanism. The WaveNet architecture achieves adaptability to diverse noise types through its stack of dilated causal convolutions with exponentially increasing dilation rates ($1, 2, 4, 8 \times 2$ blocks). This design creates a large effective receptive field that captures both short-range artifacts (e.g., high-frequency EMG bursts) and long-range correlations (e.g., slow-drift power-line interference at 50/60 Hz) without requiring explicit noise-type classification. During training, the model is exposed to a mixture of Gaussian, power-line, and EMG noise at varying SNR levels in $\{0, 5, 10, 20\}$ dB, enabling the learned filters to implicitly distinguish and suppress heterogeneous noise patterns in a data-driven manner.

Noise synthesis and denoiser training. To train Stage I, we construct paired samples $(x_{\text{noisy}}, x_{\text{clean}})$ using only training epochs from the known-subject split. We define x_{clean} as the pseudo-clean EEG from the dataset and generate $x_{\text{noisy}} = x_{\text{clean}} + \alpha n$, where n is sampled from Gaussian / power-line / EMG noise and α is set to match SNR levels in $\{0, 5, 10, 20\}$ dB. Care is taken to ensure no information leakage between training and evaluation splits during noise mixing. The WaveNet denoiser is optimized with the SI-SNR objective to recover x_{clean} from x_{noisy} .

B. Stage II: Discriminative Representation Learning

In the second stage, denoised signals are processed by a deep metric learning model to learn identity-discriminative embeddings [5], [6]. Residual convolutional neural networks (ResNet) are used as backbone architectures due to their stability and strong representation capacity [11]. Angular margin–based loss functions are employed to enforce compact intra-class clustering and clear inter-class separation in the embedding space [7]. In addition, general pair-weighted metric objectives (e.g., multi-similarity loss) can further improve embedding discrimination by leveraging multiple positive and negative relationships within a batch [8]. This embedding-based formulation naturally supports open-set recognition, enabling the detection of unseen users based on distance to learned identity prototypes in the embedding space [5], [6].

IV. EXPERIMENTAL SETUP

Dataset. We used the publicly available *Auditory Evoked Potential EEG–Biometric Dataset* from PhysioNet (2024), collected at the University of Chieti–Pescara, Italy [4]. It contains EEG recordings from 20 participants over 12 sessions of approximately two minutes each, covering resting-state (eyes

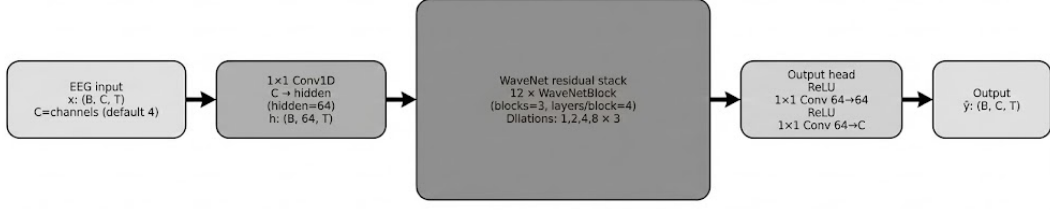


Fig. 2. WaveNetDenoiser architecture for EEG denoising.

open/closed) and auditory-stimulation conditions. During auditory stimulation, subjects listened to native-language speech, foreign-language speech, and neutral sounds through both in-ear and bone-conduction headphones. The multiple sessions, stimulus types, and headphone modalities introduce inter-session and acquisition variability, making the dataset suitable for evaluating noise robustness and open-set authentication.

Protocol and baselines. EEG recordings were segmented into fixed-length epochs and divided into training and test sets using a subject-independent protocol, ensuring that test identities were not seen during training. We compared our proposed two-stage framework against several deep learning baselines, including recurrent models (LSTM and BiLSTM) [9], [10] and single-stage convolutional architectures trained on raw signals without explicit denoising [1]–[3]. All methods were trained under identical conditions to ensure fair comparison. Performance was evaluated using Precision@1 (P@1) for identity retrieval and TAR (%) for open-set acceptance of known users [5], [6].

Open-set protocol (subject-independent). We adopt a subject-independent open-set evaluation in which the 20 subjects are partitioned into two disjoint sets: $\mathcal{S}_{\text{known}} \cap \mathcal{S}_{\text{unknown}} = \emptyset$. The embedding network $f(\cdot)$ is trained only on epochs from $\mathcal{S}_{\text{known}}$. Specifically, 16 subjects (80%) form the known set $\mathcal{S}_{\text{known}}$, while the remaining 4 subjects (20%) form the unknown set $\mathcal{S}_{\text{unknown}}$. The unknown subjects are selected randomly with a fixed seed to ensure reproducibility. For each known subject, epochs are split with a stratified shuffle split into training (64%), validation (16%), and test (20%) subsets. Training epochs are used to optimize the denoiser and embedding network, validation epochs are used for hyperparameter tuning and threshold calibration, and test epochs are used as genuine queries. We denote the held-out query epochs from known and unknown subjects by $\mathcal{Q}_{\text{known}}$ and $\mathcal{Q}_{\text{unknown}}$, respectively. Queries from $\mathcal{Q}_{\text{unknown}}$ act as impostor samples and should be rejected by the system.

Enrollment. We set $\mathcal{S}_{\text{enroll}} = \mathcal{S}_{\text{known}}$ and compute a prototype for each enrolled subject c using training/enrollment epochs $\mathcal{E}_c^{\text{train}}$: $\mu_c = \frac{1}{|\mathcal{E}_c^{\text{train}}|} \sum_{x \in \mathcal{E}_c^{\text{train}}} f(x)$. Held-out test epochs from known subjects are matched against these prototypes,

while epochs from $\mathcal{S}_{\text{unknown}}$ remain excluded from enrollment.

Decision rule. For a query x , we compute $z = f(x)$ and $d^* = \min_{c \in \mathcal{S}_{\text{enroll}}} \|z - \mu_c\|_2$. We accept the closest identity if $d^* \leq \tau$; otherwise we reject as unknown.

Threshold calibration. The threshold τ is calibrated without using test queries, using a held-out calibration split of enrolled subjects (e.g., separate sessions/epochs). We select τ to satisfy a target False Acceptance Rate (FAR) of 1% on the calibration set. The FAR is defined as the fraction of impostor queries from the unknown-subject split incorrectly accepted: $\text{FAR}(\tau) = |\{x \in \mathcal{Q}_{\text{unknown}} : d^*(x) \leq \tau\}| / |\mathcal{Q}_{\text{unknown}}|$. The corresponding False Rejection Rate (FRR) at this operating point is $\text{FRR}(\tau) = 1 - \text{TAR}(\tau)$. For example, at $\text{TAR}@1\%\text{FAR} \approx 95.64\%$ under power-line noise, the FRR is approximately 4.36%, indicating that only a small fraction of legitimate users are incorrectly rejected. Once selected, τ is fixed for all test evaluations; each model is calibrated with the same procedure for fairness.

V. RESULTS AND DISCUSSION



Fig. 3. Performance comparison between single-stage recurrent baselines (LSTM and BiLSTM).

Experimental results show that the proposed two-stage framework improves retrieval performance over recurrent baselines and is competitive with or better than the single-stage ResNet baseline across the tested noise conditions. As illustrated in Fig. 3, recurrent models (LSTM/BiLSTM) provide competitive performance under clean conditions. Still, their reliability degrades under noise and session variability, which is consistent with the known sensitivity of temporal models to distribution shifts and corrupted inputs [9], [10].

Comparison with baseline methods. We compare our proposed two-stage framework against three baseline approaches: LSTM, BiLSTM, and Single-Stage ResNet-34 (trained directly on noisy data without enhancement). As shown in Table I, the recurrent baselines (LSTM and BiLSTM) achieve reasonable accuracy in low-noise scenarios but their performance drops sharply as noise complexity increases. Under EMG noise, LSTM achieves P@1 of 87.10% and BiLSTM achieves 88.50%, while our proposed method achieves 97.62%—an improvement of over 9 percentage points. The Single-Stage ResNet-34 (without denoising) provides a stronger baseline, achieving P@1 of 96.66% under EMG noise. By integrating the WaveNet denoiser (Two-Stage), performance improves to 97.62% (+0.96%). Open-set robustness (TAR@1%FAR) improves from 95.25% to 95.51% under EMG noise and from 95.38% to 95.64% under power-line noise, while remaining essentially unchanged under Gaussian noise (95.37% vs. 95.38%). These results suggest that explicit signal enhancement is most beneficial in harder noise regimes while preserving comparable open-set behavior under milder Gaussian corruption.

TABLE I
COMPARISON WITH BASELINE METHODS (P@1 AND TAR@1%FAR).

Noise	Method	P@1 (%)	TAR@1%FAR (%)
Gaussian	LSTM (Single-stage)	89.21	88.45
Gaussian	BiLSTM (Single-stage)	90.15	89.12
Gaussian	ResNet-34 (Single-stage)	93.31	95.38
Gaussian	Proposed (Two-stage)	93.80	95.37
Power-line	LSTM (Single-stage)	85.40	82.10
Power-line	BiLSTM (Single-stage)	86.75	83.50
Power-line	ResNet-34 (Single-stage)	96.92	95.38
Power-line	Proposed (Two-stage)	98.15	95.64
EMG	LSTM (Single-stage)	87.10	85.30
EMG	BiLSTM (Single-stage)	88.50	86.20
EMG	ResNet-34 (Single-stage)	96.66	95.25
EMG	Proposed (Two-stage)	97.62	95.51

A. Effect of Metric Learning Objectives

Table II compares different deep metric learning objectives on ResNet backbones. Across architectures, metric learning losses significantly improve discriminative embeddings compared with conventional closed-set formulations by explicitly optimizing distances in the embedding space [5], [6]. In particular, multi-similarity loss and ArcFace achieve the strongest performance, indicating better intra-class compactness and inter-class separability [7], [8]. ResNet backbones are used due to their stable optimization and strong representational capacity [11].

Overall, the proposed approach maintains high identity retrieval performance (P@1 up to 98.84%) and achieves strong open-set verification performance (TAR@1%FAR up to 95.64%), demonstrating robust generalization beyond the closed-set assumption commonly adopted in earlier EEG biometric systems [1]–[3].

B. Robustness Under Noise and Ablation Study

Table III reports P@1 and SI-SNR across different noise types. The results indicate that the framework remains stable

TABLE II
PERFORMANCE COMPARISON OF DEEP METRIC LEARNING LOSSES ON RESNET BACKBONES.

Backbone	Loss Function	Accuracy (%)
ResNet-18	Triplet Loss	97.89
ResNet-18	Contrastive Loss	97.96
ResNet-18	Multi-Similarity	98.67
ResNet-18	ArcFace	98.60
ResNet-34	Triplet Loss	94.27
ResNet-34	Contrastive Loss	98.07
ResNet-34	Multi-Similarity	98.84
ResNet-34	ArcFace	98.81
ResNet-50	Triplet Loss	92.76
ResNet-50	Contrastive Loss	97.65
ResNet-50	Multi-Similarity	98.52
ResNet-50	ArcFace	98.56

under Gaussian, power-line, and EMG noise, suggesting that explicit enhancement improves downstream embedding discrimination. (Note that SI-SNR is included as a reconstruction-oriented indicator of denoising quality [12].)

TABLE III
PERFORMANCE OF THE PROPOSED TWO-STAGE FRAMEWORK UNDER DIFFERENT NOISE CONDITIONS.

Noise	Model	P@1	SI-SNR (dB)
Gaussian	ResNet-34 + Multi-Similarity	0.9380	12.62
Gaussian	ResNet-18 + Multi-Similarity	0.9424	12.61
Gaussian	ResNet-34 + ArcFace	0.9336	12.69
Power-line	ResNet-34 + Multi-Similarity	0.9815	35.55
Power-line	ResNet-18 + Multi-Similarity	0.9842	34.09
Power-line	ResNet-34 + ArcFace	0.9806	35.89
EMG	ResNet-34 + Multi-Similarity	0.9762	14.18
EMG	ResNet-18 + Multi-Similarity	0.9626	14.17
EMG	ResNet-34 + ArcFace	0.9644	14.09

To quantify the contribution of Stage I, Table IV presents an ablation study comparing single-stage training versus the full two-stage pipeline under each noise condition. Removing the noise-adaptive enhancement module results in a consistent drop in P@1, indicating that denoising improves embedding stability and helps preserve the discriminative structure for unseen-user detection. The Gaussian-noise ablation also shows a modest TAR decrease, suggesting that threshold calibration under relatively mild corruption remains an area for further refinement. These findings support the hypothesis that explicitly separating noise mitigation from representation learning yields a more robust authentication pipeline in challenging conditions.

TABLE IV
ABLATION STUDY OF THE PROPOSED TWO-STAGE FRAMEWORK UNDER DIFFERENT NOISE CONDITIONS.

Noise	Framework	Model	P@1	TAR (%)
Gaussian	Single-stage	ResNet-34 + Multi-Similarity	0.9331	95.38
Gaussian	Two-stage	ResNet-34 + Multi-Similarity	0.9380	94.37
Power-line	Single-stage	ResNet-34 + Multi-Similarity	0.9692	95.38
Power-line	Two-stage	ResNet-34 + Multi-Similarity	0.9815	95.64
EMG	Single-stage	ResNet-34 + Multi-Similarity	0.9666	95.25
EMG	Two-stage	ResNet-34 + Multi-Similarity	0.9762	95.51

From a system-level perspective, the proposed two-stage design enables targeted optimization of enhancement and

embedding objectives, improving interpretability and robustness in realistic conditions. This is particularly important for EEG-based authentication, where session variability and non-stationary noise frequently degrade performance [1], [2], [4].

C. Computational Cost and Latency

For practical deployment in real-time authentication scenarios, computational efficiency is an important consideration. Table V summarizes the model complexity and inference latency of each stage measured on a single NVIDIA RTX 3060 GPU with 4-second input epochs.

TABLE V
COMPUTATIONAL COST OF EACH STAGE.

Module	Params	FLOPs	Latency (ms)
Stage I: WaveNet Denoiser	~2.1M	~0.8G	~12
Stage II: ResNet-34 + Head	~21.3M	~3.6G	~8
Total (end-to-end)	~23.4M	~4.4G	~20

The total end-to-end inference latency for processing a single 4-second query epoch is approximately 20 ms, well within the requirements for real-time authentication (typically <500 ms). The WaveNet denoiser accounts for roughly 60% of the inference time despite having fewer parameters, due to its sequential dilated convolution structure. The total model size of ~23.4M parameters is comparable to standard mobile-deployable models. These results suggest that the two-stage framework is feasible for deployment on GPU-equipped edge devices, though further optimization (e.g., model pruning, quantization) would be beneficial for resource-constrained wearable platforms.

D. Limitations

Several limitations should be acknowledged. First, the evaluation is conducted on a single dataset with 20 subjects; generalization to larger populations and different EEG acquisition hardware remains to be validated. Second, the noise synthesis procedure uses additive noise at fixed SNR levels, which may not fully capture the complexity of real-world noise mixtures. Third, while the end-to-end latency is suitable for GPU-equipped devices, deployment on ultra-low-power wearable platforms would require model compression techniques not explored in this work. Fourth, the comparison focuses on strong deep learning baselines; incorporating handcrafted or more classical EEG biometric baselines would further contextualize the observed gains. Finally, the 4-subject unknown set may be insufficient to capture the full diversity of impostor patterns in practical scenarios.

VI. CONCLUSION

This research presented a two-stage intelligent authentication framework that integrates adaptive noise mitigation with deep metric learning-based representation learning. By explicitly decoupling denoising from identity discrimination, the proposed approach enhances robustness, scalability, and open-set recognition. Experimental results on the Auditory

Evoked Potential EEG–Biometric Dataset (20 subjects, 12 sessions) demonstrate that the framework outperforms recurrent baselines and remains competitive with or better than a single-stage ResNet baseline, achieving P@1 up to 98.15% and TAR@1%FAR up to 95.64% across three noise conditions. The ablation study confirms that integrating noise-adaptive enhancement consistently improves closed-set retrieval and strengthens open-set verification under power-line and EMG noise, while Gaussian-noise TAR remains comparable. With an end-to-end inference latency of approximately 20 ms, the framework appears suitable for real-time applications on GPU-equipped devices. However, the current evaluation is limited to a single dataset of 20 subjects; future work should validate generalization across larger and more heterogeneous populations. Overall, the results suggest that the proposed design is a practical direction for secure authentication and can be extended to other intelligent sensing and biometric applications.

VII. FUTURE WORK

Future work will validate the framework on larger and more heterogeneous EEG datasets and investigate stronger denoising backbones, including state-space sequence models, for improved robustness under non-stationary artifacts.

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